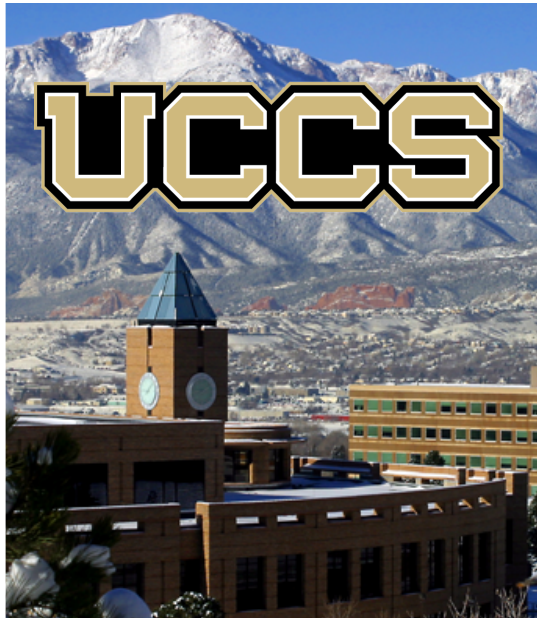




Symmetric Cryptography

Classical Cipher: Transposition

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Module Objectives:

Classical Cipher: Transposition

Transposition Cipher, e.g.,
Rail-Fence and Transposition

Transposition Cipher Security

Product Cipher

Transposition Cipher

Re-arrange the order/positions of the alphabets without altering their values

Rail Fence Cipher

List the plaintext alphabets diagonally over a number of rows, and then retrieve alphabets row by row

Rail Fence Cipher (3 Rows)

List the plaintext alphabets diagonally over a number of rows, and then retrieve alphabets row by row

E.g., M M T
 E T E A E
 E L R

The diagram shows the plaintext 'METAL R' being written diagonally across three rows. The first row contains 'M', 'M', and 'T'. The second row contains 'E', 'T', 'E', 'A', and 'E'. The third row contains 'E', 'L', and 'R'. Arrows indicate the diagonal path: one arrow points from the 'M' in the first row down to the 'E' in the second row, and another arrow points from the 'T' in the second row up to the 'M' in the first row, illustrating the zigzag pattern.

Rail Fence Cipher (3 Rows)

List the plaintext alphabets diagonally over a number of rows, and then retrieve alphabets row by row

E.g.,

M		M		T	
	E		T		E
		E		A	
			E		L
				E	
					R

MMT
ETEAE
ELR

Rail Fence Cipher (3 Rows)

List the plaintext alphabets diagonally over a number of rows, and then retrieve alphabets row by row

Ciphertext: MMTETEAELR

E.g.,

M		M		T		
	E		T		E	A
		E				L
						R

MMT
ETEAE
ELR

Permutation Cipher

List the plaintext alphabets row by row
and retrieve the ciphertext alphabets
column by column

Key determines the column order and is
a permutation of a set of size n

Key length (n) corresponds to n columns

Permutation Cipher Example

Key = [4 3 1 2] // Key length specifies
the number of columns

M	E	E	T
---	---	---	---

M	E	L	A
---	---	---	---

T	E	R	
---	---	---	--

Permutation Cipher Example

Key = [4 3 1 2] // Key length specifies
the number of columns

M	E	E	T
---	---	---	---

M	E	L	A
---	---	---	---

T	E	R	x
---	---	---	---

// Can also fill with
arbitrary alphabets

Permutation Cipher Example

Key = [4 3 1 2] // Key length specifies
 $n=4$ the number of columns

M E E T

M E L A

T E R

Permutation Cipher Example

Key = [4 3 1 2] // Key length specifies
 $n=4$ the number of columns

M E E T

M E

T E

If Key is n alphabets long,
there are $n!$ possible keys
where $n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$

Permutation Cipher Example

Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	



Ciphertext: ELR

Permutation Cipher Example

Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	



Ciphertext: ELRTA

Permutation Cipher Example

Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	



Ciphertext: ELRTAEEE

Permutation Cipher Example

Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	



Ciphertext: ELRTAEEEMMT

Permutation Cipher and Transposition Cipher

Any transposition cipher can be generalized by a permutation cipher with a key of length equal to the plaintext

One row in the matrix in this case

Transposition Cipher Security

The alphabet values do not change

=> The frequency distribution is the same

Transposition Cipher Security

The alphabet values do not change

=> The frequency distribution is the same

Vulnerable to cryptanalysis,

e.g., known/chosen plaintext attack

Permutation Cipher Example

Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	

Ciphertext: ELRTAEEEMMT

Known Plaintext Attack



Key = [? ? ? ?]

M	E	E	T
M	E	L	A
T	E	R	

“Where does MMT occur?”

Ciphertext: ELRTAEEEMMT

Known Plaintext Attack



Key = [4 ? ? ?]

M	E	E	T
M	E	L	A
T	E	R	

“Where does MMT occur?”

Ciphertext: ELRTAEEEMMT

Known Plaintext Attack



Key = [4 3 ? ?]

M	E	E	T
M	E	L	A
T	E	R	

Ciphertext: ELRTAEEEMIMT

Known Plaintext Attack



Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	

Ciphertext: ELRTAEEMMT

Known Plaintext Attack



Key = [4 3 1 2]

M	E	E	T
M	E	L	A
T	E	R	?
...	...		

ELR...TA?...EEE...MMT...

Chosen Plaintext Attack



Key = [? ? ? ?]

X	Y	Z	A	Try chosen plaintext
B	C	D	E	XYZABCDEFGHI
F	G	H	I	

Ciphertext: ZDHA EIYCGXBF

Chosen Plaintext Attack



Key = [4 3 1 2]

X	Y	Z	A	Try chosen plaintext
B	C	D	E	<u>XYZ</u> ABCDEFGHI
F	G	H	I	

Ciphertext: ZDHAEIYCGXBF

Product Cipher

Combinations of substitution ciphers
and transposition ciphers in succession

Improve security

Modern ciphers use product ciphers

